Marginal Treatment Effect

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Building bridges (Heckman, JEL 2010)

Heckman (2010) sees a major problem in the approach to program evaluation as promoted by a paradigm of RCT, that defines causality as the result of an ideal experiment.

There is a conflation of separated tasks:

- Defining causality;
- Identifying causal parameters.

This is the consequence of an absence of economic model, which led to the abandonment of distinctions between ...

- *ex ante* and *ex post* outcomes,
- subjective and objective evaluations.

Building bridges (Heckman, JEL 2010)

Policy analysis:

- all about identifying counterfactual states, that is possible outcomes in different hypothetical states of the world;
- example: the set of outcomes of the same persons in different tax regimes.

Causal comparisons:

- contrasts between outcomes in alternative possible states,
- holding factors other than the features of the policy being analyzed the same across the contrasts;
- the person subject to a particular policy is the same as the person who is not,
- except for treatment status and, possibly, the outcome associated with treatment status.

Building bridges (Heckman, JEL 2010)

Defining counterfactual is where the economic theory is needed:

- Descriptions of hypothetical worlds obtained by varying - hypothetically - the factors determining outcomes.
- may be motivated by empirical evidence,
- however, abstract representations of the evidence with an internal logic of their own.

Identifying causal parameters from data is where mathematical and statistical tools of estimation and test theory are needed:

- identification problem,
- solving practical problems of inference from empirical samples.

Building bridges (Heckman, JEL 2010)

Heckman identifies three types of problems in a policy evaluation exercise.

P1 *"Evaluating the Impacts of Implemented Interventions on Outcomes Including Their Impacts on the Well-Being of the Treated and Society at Large"*

P2 *"Forecasting the Impacts (Constructing Counterfactual States) of Interventions Implemented in One Environment in Other Environments, including Impacts on Well-Being."*

P3 *"Forecasting the Impacts of Interventions (Constructing Counterfactual States Associated with Interventions) Never Historically Experienced, including Their Impacts on Well-Being."*

What can we do so far

Ideally, want to learn individual treatment effects but we can't: fundamental problem of causal inference!

Barring that, want to learn distribution of treatment effects, but we can't: fundamental problem of causal inference!

ATE (or conditional ATE) usually considered best we can do. Identified by "gold standard" placebo controlled, randomized trial with perfect compliance. But...

- Even when treatment is randomly assigned, can't force people to take it: *randomized encouragement design*.
- Intent-to-treat (ITT) effect: causal effect of offering treatment. "Diluted" by people offered who don't take (typically assume exclusion restriction).

What can we do so far

IV allows us to go beyond ITT effects, but if treatment effects are heterogeneous, we recover the **LATE: average effect for compliers**.

Is the LATE an interesting quantity? In some contexts it is the answer to **P1**.

The research on Marginal Treatment Effects can be seen as an attempt to **extrapoLATE**: Go "beyond LATE" to more interesting causal parameters.

That is the topic of this lecture.

Beyond LATE in a "Textbook" Model

$$
\begin{aligned}
Y_0 &= \mu_0 + U_0 \\
Y_1 &= \mu_1 + U_1 \\
D &= 1\{\gamma_0 + \gamma_1 Z > V\} \\
Y &= (1-D)Y_0 + DY_1
\end{aligned}
$$

$$
Z \sim \text{Bernoulli}(q) \quad \perp (V, U_0, U_1)
$$

$$
\begin{pmatrix} V \\ U_0 \\ U_1 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_0\rho_0 & \sigma_1\rho_1 \\ \sigma_0\rho_0 & \sigma_0^2 & \sigma_{01} \\ \sigma_1\rho_1 & \sigma_{01} & \sigma_1^2 \end{pmatrix} \right)
$$

- Heckman, Tobias & Vytlacil (2001), Angrist (2004)
- Treatment effects $Y_1 - Y_0$ are heterogeneous
- $ATE = \mu_1 - \mu_0$

Selection into treatment depends on:

1. Binary instrument / encouragement Z
2. Heterogeneous cost / resistance V

Simulation

For the values , please $\mu_1 = \mu_0 = 0$, $\sigma_0 = \sigma_1 = 1$, $\sigma_{01} = 1/2$, $\rho_0 = 0.5$, $\rho_1 = 0.2$:

1- simulate 5,000 observations setting the seed: 1983.

2- The first observations of Y_1, Y_0, Δ.

3- Produce an histogram of Δ.

4- Produce a scatter Plot of (V, Δ).

Simulation: $\mu_1 = \mu_0 = 0$, $\sigma_0 = \sigma_1 = 1$, $\sigma_{01} = 1/2$

```r
library(mvtnorm)
library(tidyverse)

rho0 <- 0.5
rho1 <- 0.2

S <- matrix(c(
  1,     rho0, rho1,
  rho0,  1,    0.5,
  rho1,  0.5,  1
), 3, 3, byrow = TRUE)

set.seed(1983)

sims <- rmvnorm(5e3, sigma = S)
colnames(sims) <- c("V", "Y0", "Y1")
```

Simulation output

```r
# show a small printed sample similar to the slide
knitr::kable(head(sims, 10), digits = 3, caption = "First 10 rows of sims")
```

Table 1: First 10 rows of sims

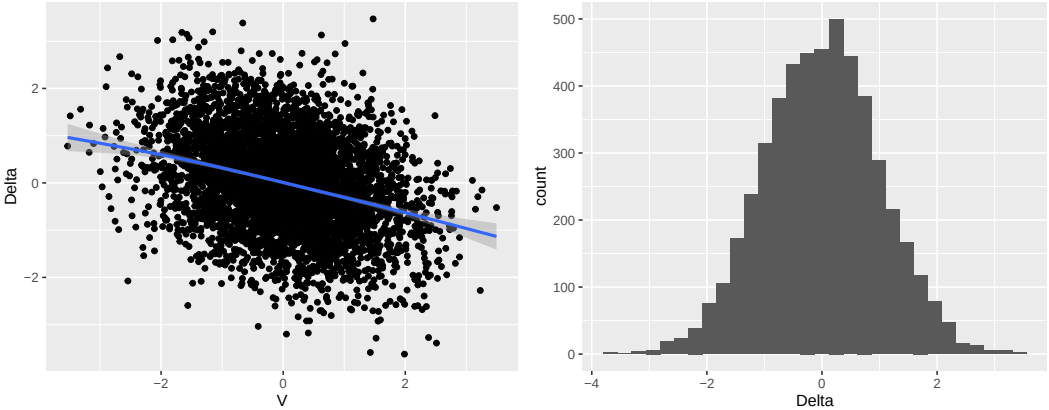
V	Y0	Y1	Delta
-0.122	-0.399	1.079	1.478
-0.506	-1.104	1.487	2.592
0.005	-0.121	-0.456	-0.335
-0.549	-0.248	-0.899	-0.651
1.950	-0.095	-0.675	-0.580
0.561	0.112	-0.615	-0.726
-0.238	-0.439	-1.535	-1.096
-1.461	-1.225	-0.055	1.170
-0.336	-0.891	1.531	2.422
-0.845	-0.274	0.637	0.911

```r
library(ggplot2)


# Scatter: V vs Delta with a smoothed line
DV_scatter <- sims |>
ggplot(aes(x = V, y = Delta)) +
geom_point() +
geom_smooth()

Dhist <- sims |>
ggplot(aes(x = Delta)) +
geom_histogram()
```

```r
library(gridExtra)
# Arrange side-by-side (like the slide)
grid.arrange(DV_scatter, Dhist, ncol = 2)
```



Propoerties of the "Textbook" Model

$$
\begin{aligned}
Y_0 &= \mu_0 + U_0 \\
Y_1 &= \mu_1 + U_1 \\
D &= 1\{\gamma_0 + \gamma_1 Z > V\} \\
Y &= (1-D)Y_0 + DY_1
\end{aligned}
$$

$$
Z \sim \text{Bernoulli}(q) \quad \perp (V, U_0, U_1)
$$

$$
\begin{pmatrix} V \\ U_0 \\ U_1 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_0\rho_0 & \sigma_1\rho_1 \\ \sigma_0\rho_0 & \sigma_0^2 & \sigma_{01} \\ \sigma_1\rho_1 & \sigma_{01} & \sigma_1^2 \end{pmatrix} \right)
$$

The treatment effects are distributed as a normal random variable.

$$
\Delta \equiv Y_1 - Y_0 \sim \mathcal{N}(\mu_1 - \mu_0,\ \sigma_0^2 + \sigma_1^2 - 2\sigma_{01})
$$

The Treatment effect depends linearly of the resistance term:

$$
\text{Cov}(\Delta_i, V_i) = \text{Cov}(Y_{1i}, V_i) - \text{Cov}(Y_{0i}, V_i) = \sigma_1\rho_1 - \sigma_0\rho_0
$$

Simulation

For values $\gamma_0 = -1$, $\gamma_1 = 1.5$:

1- Define the compliers

2- Present them on a graph

3- Compute the share of compliers and the LATE

LATE for the Textbook Model

- LATE = average effect for compliers
- Since

$$D = 1(\gamma_0 + \gamma_1 Z > V)$$

compliers are defined by

$$\gamma_0 \leq V < \gamma_0 + \gamma_1$$

```
gamma0 <- -1
gamma1 <- 1.5

sims <- sims |>
  mutate(complier = (V >= gamma0) & (V < gamma0 + gamma1))
```

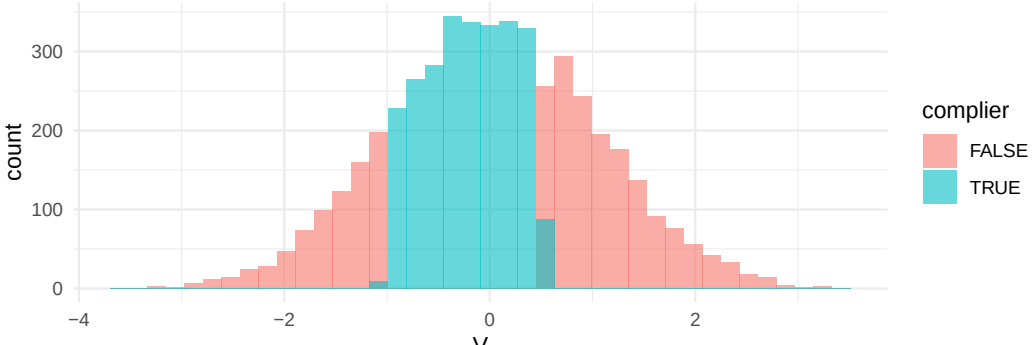
```r
head(sims)
```

```
## # A tibble: 6 x 5
##          V      Y0     Y1  Delta complier
##      <dbl>   <dbl>  <dbl>  <dbl> <lgl>
## 1 -0.122   -0.399   1.08   1.48  TRUE
## 2 -0.506   -1.10    1.49   2.59  TRUE
## 3  0.00457 -0.121  -0.456 -0.335 TRUE
## 4 -0.549   -0.248  -0.899 -0.651 TRUE
## 5  1.95    -0.0948 -0.675 -0.580 FALSE
## 6  0.561    0.112  -0.615 -0.726 FALSE
```

Who's a complier when $\gamma_0 = -1$, $\gamma_1 = 1.5$?

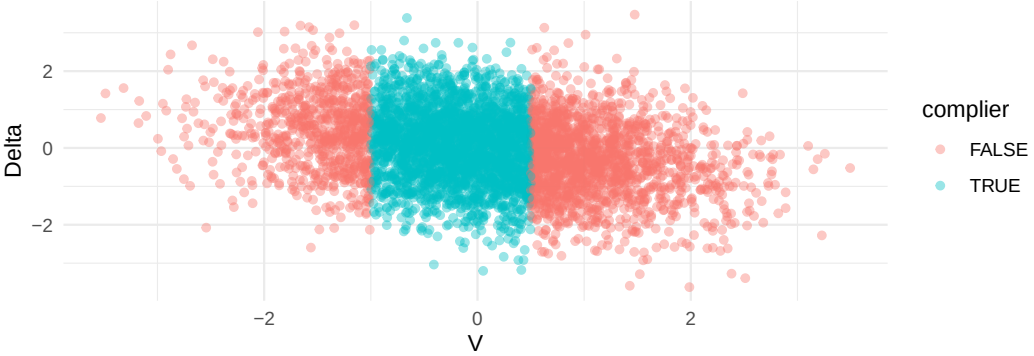
```
library(ggplot2)

ggplot(sims, aes(x = V, fill = complier)) +
  geom_histogram(bins = 40, position = "identity", alpha = 0.6) +
  labs(x = "V", y = "count") +
  theme_minimal()
```



Who's a complier when $\gamma_0 = -1$, $\gamma_1 = 1.5$?

```
ggplot(sims, aes(x = V, y = Delta, col = complier)) +
  geom_point(alpha = 0.4) +
  labs(x = "V", y = "Delta") +
  theme_minimal()
```



Who's a complier when $\gamma_0 = -1$, $\gamma_1 = 1.5$?

```r
# Share of compliers
pnorm(gamma0 + gamma1) - pnorm(gamma0)
```

```
## [1] 0.5328072
```

LATE

```r
sims |>
  group_by(complier) |>
  summarize(mean_Delta = mean(Delta)) |>
  knitr::kable(digits = 3)
```

complier	mean_Delta
FALSE	-0.083
TRUE	0.068

Different Instrument, Different LATE: $\gamma_0 = -1$, varying γ_1

Try to vary the value of γ_1 and compute the LATE for each of these values.

This is as if you had different instruments with different effect intensity (γ_1)on the individuals.

Different Instrument, Different LATE: $\gamma_0 = -1$, varying γ_1

```r
get_LATE <- function(gamma1) {
  sims |>
    mutate(complier = (V >= -1) & (V < -1 + gamma1)) |>
    filter(complier) |>
    summarize(LATE = mean(Delta)) |>
    pull()
}

gamma1_seq <- c(0.75, 1, 1.25, 1.5, 1.75, 2)

LATE_vals <- purrr::map_dbl(gamma1_seq, get_LATE)
```

Different Instrument, Different LATE: $\gamma_0 = -1$, varying γ_1

```r
rbind(gamma1_seq, LATE_vals) |>
  knitr::kable(digits = 2)
```

gamma1_seq	0.75	1.00	1.25	1.50	1.75	2
LATE_vals	0.21	0.15	0.11	0.07	0.03	0

ATT and ATNT in the Textbook Model

With the Textbook model we can compute analystically the ATT and the ATNT

$$
\begin{aligned}
\textbf{ATT} &\equiv E(\Delta \mid D=1) \\
&= E(\Delta \mid D=1, Z=0)P(Z=0 \mid D=1) + E(\Delta \mid D=1, Z=1)P(Z=1 \mid D=1) \\
&= E(\Delta \mid V<\gamma_0)\cdot(1-q_1) + E(\Delta \mid V<\gamma_0+\gamma_1)\cdot q_1
\end{aligned}
$$

$$
\begin{aligned}
\textbf{ATNT} &\equiv E(\Delta \mid D=0) \\
&= E(\Delta \mid D=0, Z=0)P(Z=0 \mid D=0) + E(\Delta \mid D=0, Z=1)P(Z=1 \mid D=0) \\
&= E(\Delta \mid V>\gamma_0)\cdot(1-q_0) + E(\Delta \mid V>\gamma_0+\gamma_1)\cdot q_0
\end{aligned}
$$

ATT and ATNT in the Textbook Model

ATT is a weighted average of $E(\Delta \mid V < \gamma_0)$ and $E(\Delta \mid V < \gamma_0 + \gamma_1)$

ATNT is a weighted average of $E(\Delta \mid V > \gamma_0)$ and $E(\Delta \mid V > \gamma_0 + \gamma_1)$

We need to compute:

$$E(\Delta \mid V > c) \quad \text{and} \quad E(\Delta \mid V < c)$$

ATT and ATNT depend on the instrument Z through γ_0 and γ_1. These define "the treated".

Can you simulate treated in the dataset?

Need Z to define "the treated"

```r
# Remove complier indicator if it exists
sims <- sims |>
  select(-any_of("complier"))

# Define instrument
set.seed(123)
q <- 0.5
sims <- sims |>
  mutate(
    Z = rbinom(nrow(sims), 1, q),
    treated = (gamma0 + gamma1 * Z > V)
  )
```

Need Z to define "the treated"

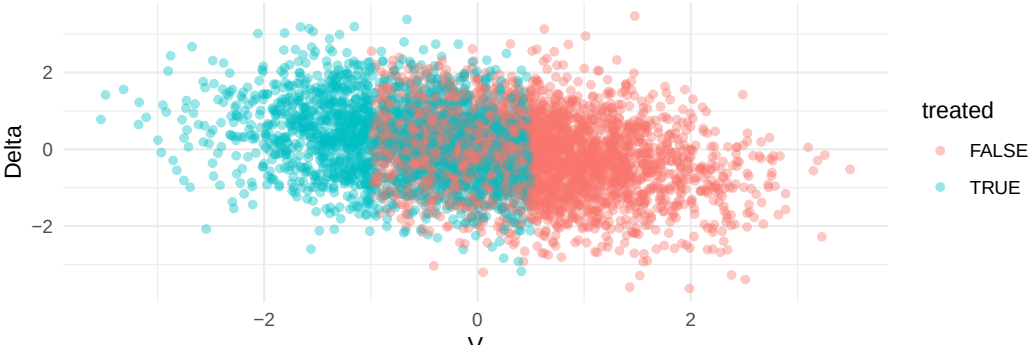
```r
head(sims)
```

```
## # A tibble: 6 x 6
##          V      Y0     Y1  Delta     Z treated
##      <dbl>   <dbl>  <dbl>  <dbl> <int> <lgl>
## 1 -0.122   -0.399   1.08   1.48      0 FALSE
## 2 -0.506   -1.10    1.49   2.59      1 TRUE
## 3  0.00457 -0.121  -0.456 -0.335     0 FALSE
## 4 -0.549   -0.248  -0.899 -0.651     1 TRUE
## 5  1.95    -0.0948 -0.675 -0.580     1 FALSE
## 6  0.561    0.112  -0.615 -0.726     0 FALSE
```

Who's treated if $q = 0.5$, $\gamma_0 = -1$, $\gamma_1 = 1.5$?

```
library(ggplot2)

ggplot(sims, aes(x = V, y = Delta, col = treated)) +
  geom_point(alpha = 0.4) +
  labs(x = "V", y = "Delta") +
  theme_minimal()
```



ATT and ATNT Effects: $q = 0.5$, $\gamma_0 = -1$, $\gamma_1 = 1.5$

```
sims |>
  group_by(treated) |>
  summarize(mean_Delta = mean(Delta)) |>
  knitr::kable(digits = 3)
```

treated	mean_Delta
FALSE	-0.165
TRUE	0.222

- Different values of q, γ_0, and γ_1 would give different ATT and ATNT.
- In this example we have **selection on gains**:

$$\text{ATT} < \text{ATE} < \text{ATNT}$$

Analytical Results for the Textbook Model

You can show that:

$$\textbf{ATE} = \mu_1 - \mu_0$$

$$\textbf{LATE} = \text{ATE} - (\sigma_1\rho_1 - \sigma_0\rho_0)\left[\frac{\phi(\gamma_0+\gamma_1) - \phi(\gamma_0)}{\Phi(\gamma_0+\gamma_1) - \Phi(\gamma_0)}\right]$$

$$\textbf{ATT} = \text{ATE} - (\sigma_1\rho_1 - \sigma_0\rho_0)\left[\frac{(1-q)\phi(\gamma_0) + q\phi(\gamma_0+\gamma_1)}{(1-q)\Phi(\gamma_0) + q\Phi(\gamma_0+\gamma_1)}\right]$$

$$\textbf{ATNT} = \text{ATE} + (\sigma_1\rho_1 - \sigma_0\rho_0)\left[\frac{(1-q)\phi(\gamma_0) + q\phi(\gamma_0+\gamma_1)}{(1-q)\{1-\Phi(\gamma_0)\} + q\{1-\Phi(\gamma_0+\gamma_1)\}}\right]$$

Example: $\sigma_0 = \sigma_1 = 1,\ q = 1/2$

Define:

$$\delta \equiv \rho_1 - \rho_0$$

In our simulation, The formulas simplify to:

$$\textbf{LATE} = -\delta \left[\frac{\phi(\gamma_0 + \gamma_1) - \phi(\gamma_0)}{\Phi(\gamma_0 + \gamma_1) - \Phi(\gamma_0)} \right]$$

$$\textbf{ATT} = -\delta \left[\frac{\phi(\gamma_0) + \phi(\gamma_0 + \gamma_1)}{\Phi(\gamma_0) + \Phi(\gamma_0 + \gamma_1)} \right]$$

$$\textbf{ATNT} = \delta \left[\frac{\phi(\gamma_0) + \phi(\gamma_0 + \gamma_1)}{\{1 - \Phi(\gamma_0)\} + \{1 - \Phi(\gamma_0 + \gamma_1)\}} \right]$$

Check the results of your simualtions against the analytical formula.

First-stage effect 0.07, $q = 1/2$, $\delta = -0.1$

```
##   Parameter       Value
## 1      LATE  0.06198937
## 2       ATT  0.20963075
## 3      ATNT -0.15498179
## 4       ATE  0.00000000
```

Why Do We Care?

In the textbook model we can see how ATE, LATE, ATT and ATNT compare.

The key parameters of the model are point identified.

This allows us to go beyond LATE using data: ATE, ATT, ATNT, and the Marginal Treatment Effect (more to come).

Marginal Treatment Effects are a modern update of the textbook model.

Heckman Two-step Estimator

With an instrument, it is always possible to apply the textbook model to your data. Esstimation proceedss with the so-called **Heckman Two-step Estimator**.

We can show that:

$$E[Y|D=1, Z=z] = \mu_1 + \delta_1 E(V|D=1, Z=z)$$
$$E(V|D=1, Z=z) = -\frac{\phi(\gamma_0 + \gamma_1 z)}{\Phi(\gamma_0 + \gamma_1 z)}$$

and

$$E[Y|D=0, Z=z] = \mu_0 + \delta_0 E(V|D=0, Z=z)$$
$$E(V|D=0, Z=z) = \frac{\phi(\gamma_0 + \gamma_1 z)}{1 - \Phi(\gamma_0 + \gamma_1 z)}$$

Heckman Two-step Estimator

Define:

$$\lambda(z) = \frac{\phi(\gamma_0 + \gamma_1 z)}{1 - \Phi(\gamma_0 + \gamma_1 z)}$$

$$\kappa(z) = -\frac{\phi(\gamma_0 + \gamma_1 z)}{\Phi(\gamma_0 + \gamma_1 z)}$$

Then:

- Estimate γ_0, γ_1 using D and Z.
- Regress Y on $\lambda(z)$ for $D = 0$ to obtain μ_0, δ_0.
- Regress Y on $\kappa(z)$ for $D = 1$ to obtain μ_1, δ_1.